

Module 4 Lesson 1

Modeling Data: Exponential, Logarithmic, and Quadratic Functions (7.6)

Objectives

In this lesson, we will learn how to

- graph exponential functions;
- use exponential models;
- graph logarithmic functions;
- use logarithmic models;
- graph quadratic functions;
- use quadratic models;
- determine an appropriate function for modeling data.

1. Scatter Plots and Regression Lines

Data presented visually as a set of points is called a scatter plot.

A line that best fits the data points in a scatter plot is called a regression line.

Scatter Plot

A scatter plot is a graph of data points.
Each point represents an ordered pair

$$(x, y).$$

Regression Line

A regression line is a line that best fits the data points in a scatter plot.

Scatter plots can help us decide whether a data set is best modeled by a linear, exponential, logarithmic, or quadratic function.

2. Exponential Functions

An exponential function has a variable in the exponent.

Exponential Function

The exponential function f with base b is defined by

$$f(x) = b^x,$$

where b is a positive constant other than 1:

$$b > 0, \quad b \neq 1.$$

The variable x may be any real number.

For example,

$$f(x) = 2^x$$

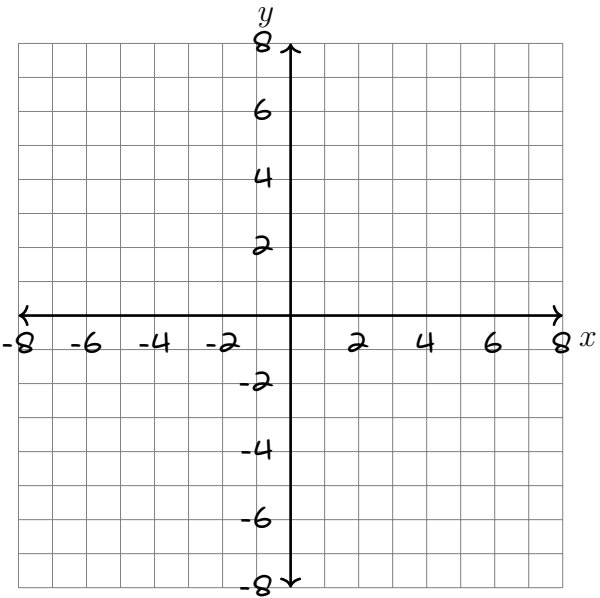
is an exponential function.

3. Graphing an Exponential Function

Example 1: Graph the exponential function

$$f(x) = 2^x.$$

x	$f(x) = 2^x$	(x, y)
-3		
-2		
-1		
0		
1		
2		
3		



Solution

Complete the table:

x	$f(x) = 2^x$	(x, y)
-3	$2^{-3} = \frac{1}{8}$	$(-3, \frac{1}{8})$
-2	$2^{-2} = \frac{1}{4}$	$(-2, \frac{1}{4})$
-1	$2^{-1} = \frac{1}{2}$	$(-1, \frac{1}{2})$
0	$2^0 = 1$	$(0, 1)$
1	$2^1 = 2$	$(1, 2)$
2	$2^2 = 4$	$(2, 4)$
3	$2^3 = 8$	$(3, 8)$

Plot the ordered pairs and connect them with a smooth curve.

4. Comparing Linear and Exponential Models

Linear and exponential models can both be used to describe real-world data.

A linear model has the form

$$f(x) = mx + b.$$

An exponential model has the form

$$g(x) = ab^x.$$

Example 2: Suppose world population, in billions, x years after 1949, is modeled by the following functions:

$$f(x) = 0.074x + 2.301$$

and

$$g(x) = 2.634(1.016)^x.$$

The linear function is f , and the exponential function is g .

- Estimate the world population in 2000. The year 2000 is 51 years after 1949.
- Compare the results with the actual world population of about 6.1 billion.
- Which function better models a projected population of 8 billion in 2026? The year 2026 is 77 years after 1949.

Solution

For the year 2000, use $x = 51$.

Linear model:

$$f(51) = 0.074(51) + 2.301.$$

$$f(51) \approx 6.1.$$

Exponential model:

$$g(51) = 2.634(1.016)^{51}.$$

$$g(51) \approx 5.9.$$

Since the actual population was about 6.1 billion, both models are close, but the linear model is closer.

For the year 2026, use $x = 77$.

$$f(77) = 0.074(77) + 2.301 \approx 8.0.$$

$$g(77) = 2.634(1.016)^{77} \approx 8.9.$$

The linear model better matches the projection of 8 billion.

5. The Natural Base e

An irrational number, symbolized by the letter e , appears as a base in many exponential functions.

$$e \approx 2.72$$

or more accurately,

$$e \approx 2.71828 \dots$$

The number e is called the natural base.

Natural Exponential Function

The function

$$f(x) = e^x$$

is called the natural exponential function.

6. Exponential Model Application

Example 3: Medical research indicates that the risk of having a car accident increases exponentially as the concentration of alcohol in the blood increases.

The risk is modeled by

$$R = 6e^{12.77x},$$

where x is the blood alcohol concentration and R , given as a percent, is the risk of having a car accident.

In many states, it is illegal to drive with a blood alcohol concentration of 0.08 or greater.

What is the risk of a car accident with a blood alcohol concentration of 0.08?

Solution

Substitute $x = 0.08$:

$$R = 6e^{12.77(0.08)}.$$

Evaluate:

$$R = 6e^{1.0216}.$$

Using a calculator,

$$R \approx 16.665813.$$

Rounded to one decimal place,

$$R \approx 16.7\%.$$

The risk of having a car accident is approximately

$$16.7\%.$$

7. Logarithmic Functions

A logarithmic function is the inverse of an exponential function.

Logarithmic Form and Exponential Form

The equation

$$y = \log_b x$$

means

$$b^y = x,$$

where

$$b > 0, \quad b \neq 1, \quad x > 0.$$

For example,

$$y = \log_2 x$$

means

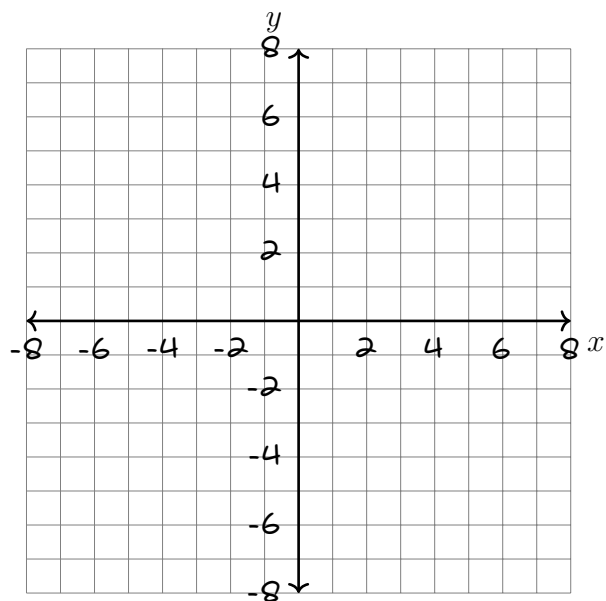
$$2^y = x.$$

8. Graphing a Logarithmic Function

Example 4: Graph the logarithmic function

$$y = \log_2 x.$$

$x = 2^y$	y	(x, y)
2^{-2}	-2	
2^{-1}	-1	
2^0	0	
2^1	1	
2^2	2	
2^3	3	



Solution

Because

$$y = \log_2 x$$

means

$$2^y = x,$$

we use exponential form to find ordered pairs.

$x = 2^y$	y	(x, y)
$2^{-2} = \frac{1}{4}$	-2	$(\frac{1}{4}, -2)$
$2^{-1} = \frac{1}{2}$	-1	$(\frac{1}{2}, -1)$
$2^0 = 1$	0	$(1, 0)$
$2^1 = 2$	1	$(2, 1)$
$2^2 = 4$	2	$(4, 2)$
$2^3 = 8$	3	$(8, 3)$

Plot the ordered pairs and connect them with a smooth curve.

9. Logarithmic Model Application

A logarithmic model often describes data that increase quickly at first and then increase more slowly.

Example 5: When the outside air temperature is between 72°F and 96°F , the temperature in an enclosed vehicle climbs rapidly at first. A logarithmic model for the temperature increase after x minutes is

$$f(x) = -11.6 + 13.4 \ln x.$$

Use the function to find the temperature increase, to the nearest degree, after 50 minutes.

Solution

Substitute $x = 50$:

$$f(50) = -11.6 + 13.4 \ln(50).$$

Using a calculator,

$$f(50) \approx 41.$$

So the temperature increase after 50 minutes is about

$$41^{\circ}.$$

This matches the actual temperature increase well.

10. Quadratic Functions

A quadratic function is any function of the form

$$y = ax^2 + bx + c$$

or

$$f(x) = ax^2 + bx + c,$$

where a , b , and c are real numbers and

$$a \neq 0.$$

Quadratic Function

A quadratic function has the form

$$f(x) = ax^2 + bx + c, \quad a \neq 0.$$

The graph of any quadratic function is called a parabola.

The vertex of a parabola is the lowest point or highest point on the graph.

11. Vertex of a Parabola

For the quadratic function

$$y = ax^2 + bx + c,$$

the x -coordinate of the vertex is

$$x = \frac{-b}{2a}.$$

After finding x , substitute it into the function to find the y -coordinate.

Vertex of a Parabola

For

$$y = ax^2 + bx + c,$$

the vertex occurs when

$$x = \frac{-b}{2a}.$$

12. Graphing a Quadratic Function

Steps for Graphing a Parabola

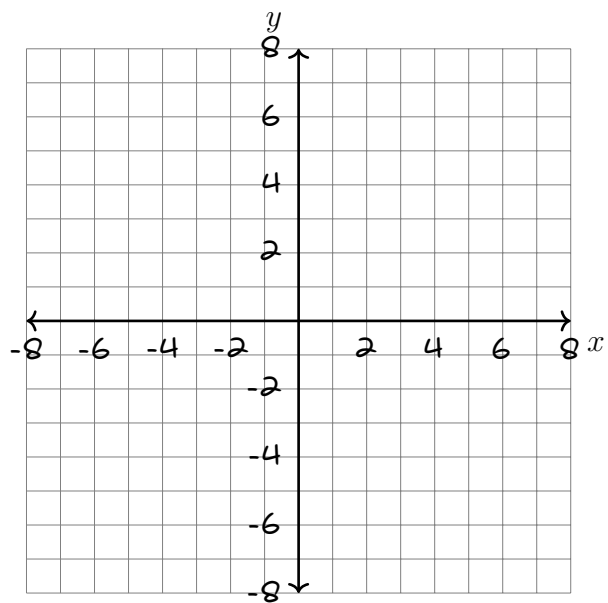
1. Determine whether the parabola opens upward or downward.
2. Find the vertex.
3. Find the x -intercepts, if they exist.
4. Find the y -intercept.
5. Plot the points.
6. Connect the points with a smooth curve.

- If $a > 0$, the parabola opens upward.
- If $a < 0$, the parabola opens downward.

13. Graphing a Parabola

Example 6: Graph the quadratic function

$$y = x^2 - 2x - 3.$$



Solution

The coefficient of x^2 is

$$a = 1.$$

Since $a > 0$, the parabola opens upward.

Find the vertex.

For

$$y = x^2 - 2x - 3,$$

we have

$$a = 1, \quad b = -2.$$

The x -coordinate of the vertex is

$$x = \frac{-b}{2a} = \frac{-(-2)}{2(1)} = 1.$$

Substitute $x = 1$ into the original function:

$$y = (1)^2 - 2(1) - 3.$$

$$y = 1 - 2 - 3 = -4.$$

The vertex is

$$(1, -4).$$

Find the x -intercepts by setting $y = 0$:

$$0 = x^2 - 2x - 3.$$

Factor:

$$0 = (x - 3)(x + 1).$$

So,

$$x - 3 = 0$$

or

$$\begin{array}{l} 15 \\ x + 1 = 0. \end{array}$$

Thus,

$$x = 3$$

or

14. Choosing an Appropriate Model

Different types of data are modeled by different types of functions.

Choosing a Model

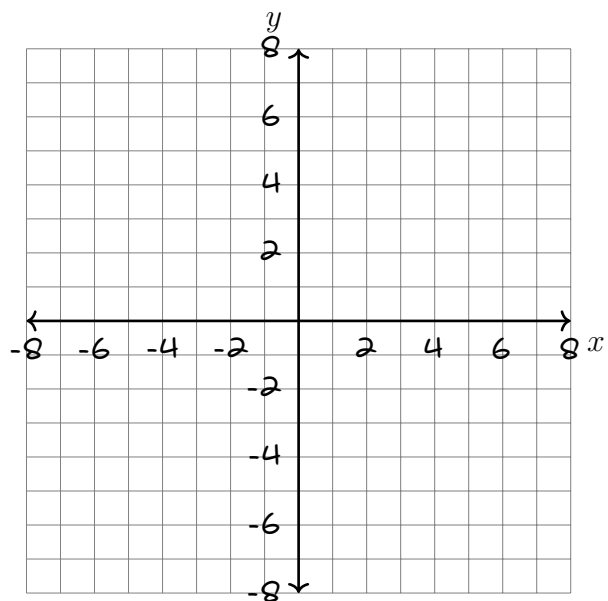
Description of Data Points	Possible Model
Lie on or near a line	Linear function $y = mx + b$
Increasing more and more rapidly	Exponential function $y = b^x$, $b > 1$
Increasing, but the rate of increase is slowing down	Logarithmic function $y = \log_b x$, $b > 1$
Decreasing and then increasing	Quadratic function $y = ax^2 + bx + c$, $a > 0$
Increasing and then decreasing	Quadratic function $y = ax^2 + bx + c$, $a < 0$

15. More Practice

Example 7: Complete the table and graph the exponential function:

$$f(x) = 3^x.$$

x	$f(x) = 3^x$	(x, y)
-2		
-1		
0		
1		
2		



Example 8: For the quadratic function

$$y = x^2 + 4x + 3,$$

find the vertex and intercepts.

Summary

Key Ideas

- A scatter plot displays data as a set of points.
- A regression line is a line that best fits data points.
- An exponential function has the form

$$f(x) = b^x, \quad b > 0, b \neq 1.$$

- The natural base is

$$e \approx 2.71828.$$

- A logarithmic equation

$$y = \log_b x$$

is equivalent to

$$b^y = x.$$

- A quadratic function has the form

$$f(x) = ax^2 + bx + c, \quad a \neq 0.$$

- The graph of a quadratic function is a parabola.
- The vertex of

$$y = ax^2 + bx + c$$

occurs when

$$x = \frac{-b}{2a}.$$

- Different patterns in scatter plots suggest different models.